

Markscheme

Mathematics: analysis and approaches
Practice question bank

144. (a)

The complex numbers $z_1 = 2 + i$ and $z_3 = 6 + 5i$ represent two opposite vertices of a square on an Argand diagram (with z_2 and z_4 the other two vertices, in order around the square).

Find the centre of the square, in the form $a + bi$.

since z_1 and z_3 are opposite vertices, the centre is the midpoint of $z_1 z_3$ (M1)

$$\text{centre} = \frac{(2 + 6) + (1 + 5)i}{2} \quad (M1)$$

$$= 4 + 3i \quad \mathbf{A1}$$

[5 marks]